Let (M^n, g) be flat $(Rm = 0)$. If in addition 1) M has infinite injectivity radius

or 2) $\mathrm{vol}\, B(p,r) \geq v r^n$ for some $v > 0$ and all $r > 0$

then $M = \mathbb{R}^n$ with $g = g_{Euc}$.

These rigidity statements lead to information about the structure of manifolds with bounded curvature: If (M, g) satisfies $|Rm| \leq K$ and either 1) $\mathrm{inj\,rad}\, p \geq i_0$

or 2) $\mathrm{vol}\, B(p,1) \geq v > 0$

then $\|(M, g, p)\|_{C^{0,\alpha}, r} \leq Q$, r, Q dep on n, K, i_0, v, α

(in fact can get $C^{1,\alpha}, r$)

To find similar results for Ricci curvature we need corresponding rigidity statements.

The splitting theorem (Cheeger - Gromoll)

Let (M^n, g) be complete, $Ric \geq 0$

Let $\gamma : (-\infty, \infty) \to M$ be a geodesic line,

i.e. all segments in γ are length minimizing

(if $|\dot\gamma| = 1$ then $d(\gamma(s), \gamma(t)) = |s-t|$)

Then $M^n = \mathbb{R} \times \tilde{M}^{n-1}$ isometrically

Corollary: if $Ric \geq 0$ and inj rad $= \infty$,

then $M = \mathbb{R}^n$ isometrically.

Bishop - Gromov volume comparison

Let (M^n, g) be complete, $Ric \geq 0$

Then $\mathrm{Vol}\, B(p,r) \leq \omega_n r^n \quad \forall r > 0$

$\omega_n =$ vol of unit ball in $\mathbb{R}^n$

with equality only if $B(p,r) = B(0,r) \subset \mathbb{R}^n$ isometrically.

These lead to corresponding structural results for spaces with bounded Ricci.

<u>Thm</u> (Anderson) $\exists\ \delta, r > 0$ (depending on n, α) satisfying the following. Let (M^n, g) satisfy $|Ric| \leq \delta$ and either

1) inj rad $p \geq 1$

or 2) Vol $B(p,1) \geq (1-\delta)\omega_n$

Then $\|(M,g,p)\|_{C^{1,\alpha},r} \leq 2$.

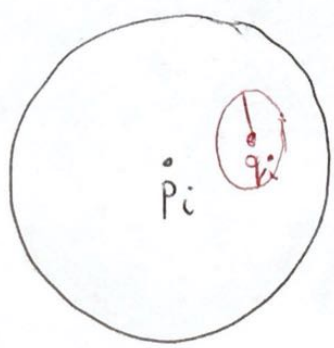
<u>Proof</u> 1) Argue by contradiction. Suppose (M_i, g_i) satisfy $|Ric(g_i)| \leq \delta_i \to 0$

inj rad $p_i \geq 1$

Define $r(q) = \sup\{r \mid \|(M_i, g_i, q)\|_{C^{1,\alpha},r} \leq 2\}$

Assume that $r(p_i) \to 0$.

Let $q_i \in B(p_i, 1)$ be



points maximizing

$$\frac{\min\{1, \text{inj rad } q\}\, d_q}{r(q)}$$

where $d_q = 1 - d(p_i, q)$

we must have $\frac{\text{inj rad } q_i}{r(q_i)} d_{q_i} = \lambda_i \to \infty$

Consider rescaled manifolds

$$(\tilde{M}_i, \tilde{g}_i, \tilde{q}_i) \qquad \begin{aligned} \tilde{M}_i &= M_i \\ \tilde{g}_i &= r(q_i)^{-1} g_i \\ \tilde{q}_i &= q_i \end{aligned}$$

<u>Key properties</u>

- $|\mathrm{Ric}\, \tilde{M}_i| \leq \delta_i \to 0$
- $r(\tilde{q}_i) = 1$
- $\text{injrad } \tilde{q}_i \to \infty$
- $\forall R > 0$, $q \in \tilde{M}_i$ s.t. $d(q_i, q) < R$ we have uniform lower bound $r(q) \geq c_R > 0$ for sufficiently large i.

<u>Note</u>: for $q \in B(q_i, \frac{1}{2} d_{q_i})$ we have

$$\frac{\text{inj rad } q}{r(q)} \leq 2 \frac{\text{inj rad } q_i}{r(q_i)}$$

So for $q \in B(\tilde{q}_i, \frac{1}{2} d_{q_i} r(q_i)^{-1})$ we have the same bound.

$$\frac{\text{inj rad } q}{r(q)} \leq 2 \frac{\text{injrad } \tilde{q}_i}{\cancel{r(\tilde{q}_i)} = 1}$$

Need to argue that injrad q is not much smaller than injrad $\tilde{q}_i$

Using these we can pass to a limit $(M_\infty, g_\infty, q_\infty)$ in pointed $C^{1,\beta}$-top. for $\beta < \alpha$.

We can work in harmonic coordinates for $\tilde{g}_i$ of a uniform size:

$$\frac{1}{2}\Delta \tilde{g}_{i,kl} = -\mathrm{Ric}(\tilde{g}_i)_{kl} + Q_{kl}(\tilde{g}_i, \partial \tilde{g}_i)$$

as $i \to \infty$ we get a weak solution of

$$\frac{1}{2}\Delta g_{\infty,kl} = Q_{kl}(\tilde{g}_\infty, \partial \tilde{g}_\infty)$$

$\Rightarrow$ g_∞ is smooth and Ricci flat.

In addition inj rad $M_\infty = \infty$ $\Rightarrow (M_\infty, q_\infty) = (\mathbb{R}^n, 0)$

On $\mathbb{R}^n$ $r(0) = \infty$.

One can show that then $r(q_i) \to \infty$ as $i \to \infty$.

This contradicts that $r(q_i) = 1$. $\square$